

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 942+650

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

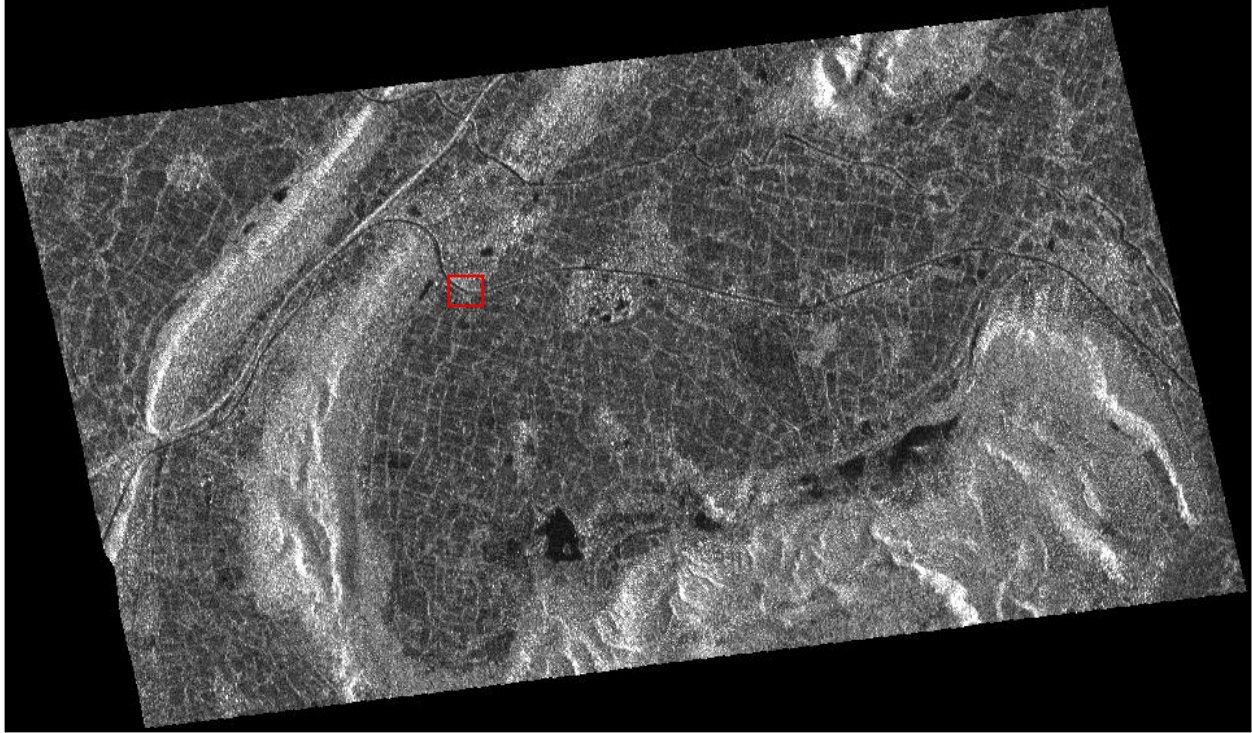
Flight Direction - DESCENDING

Polarization – VV+VH

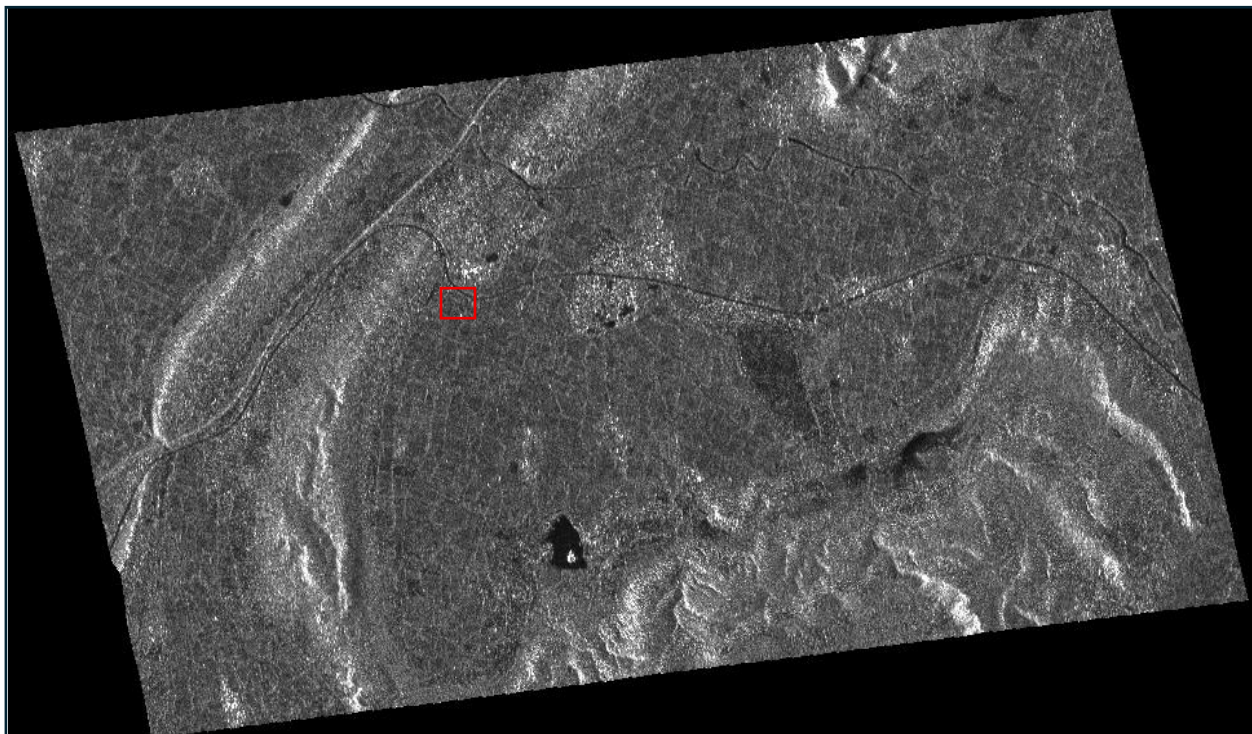
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

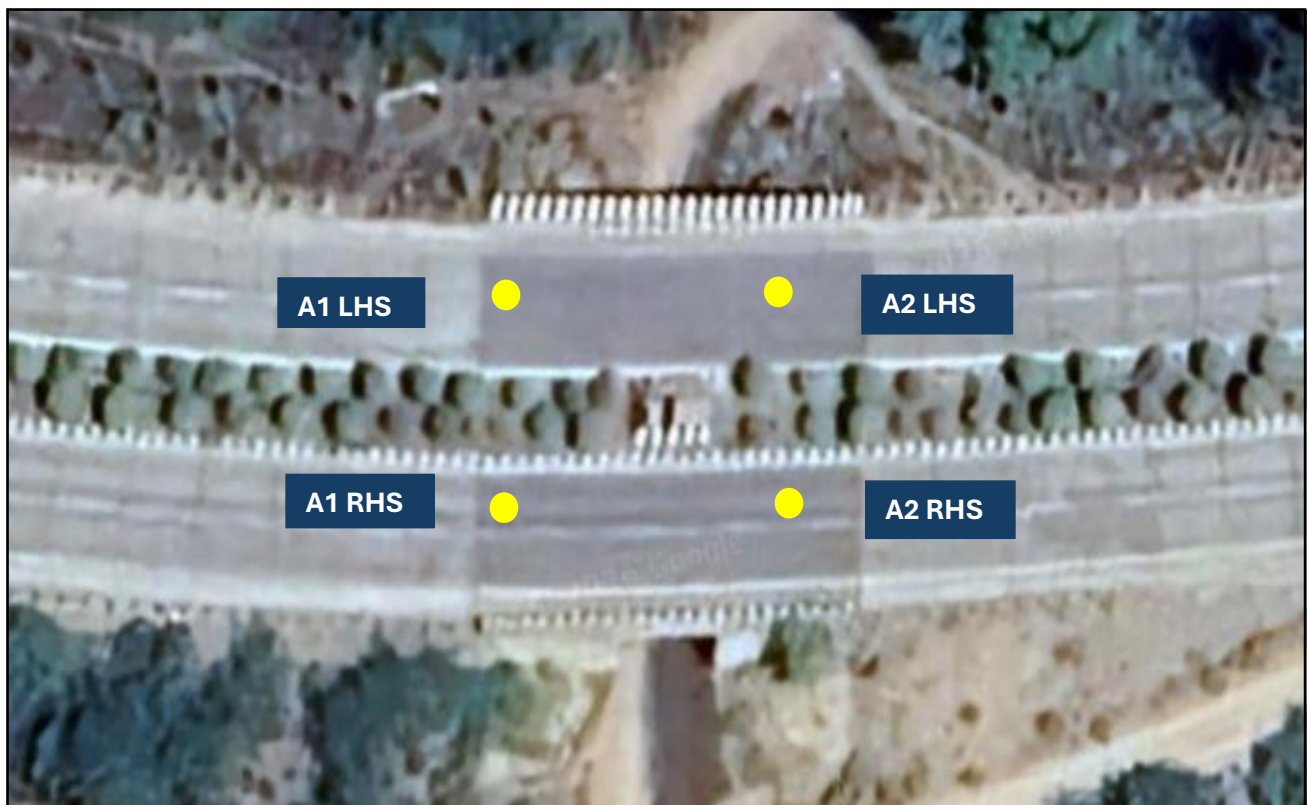


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

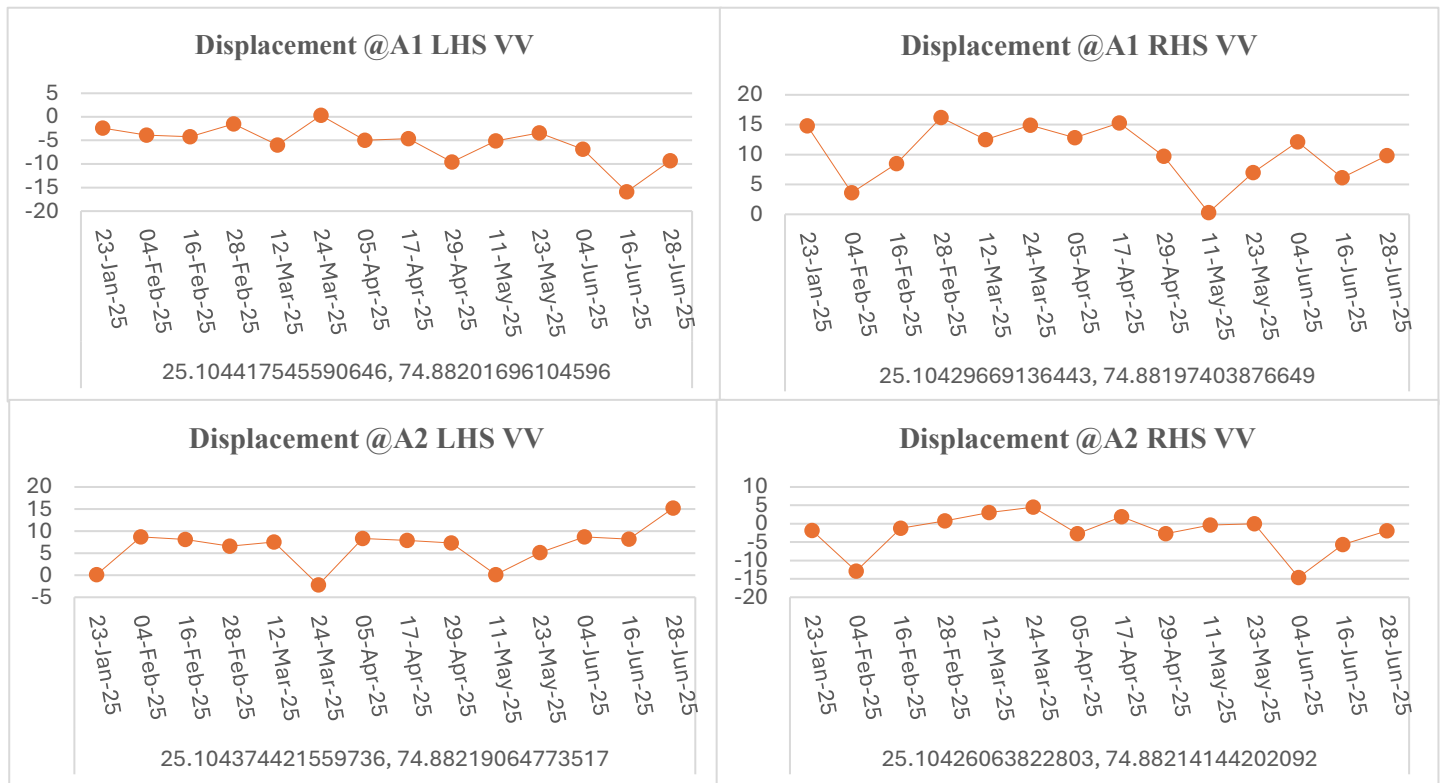
VUP – CH-942+650

Length – 22.92M

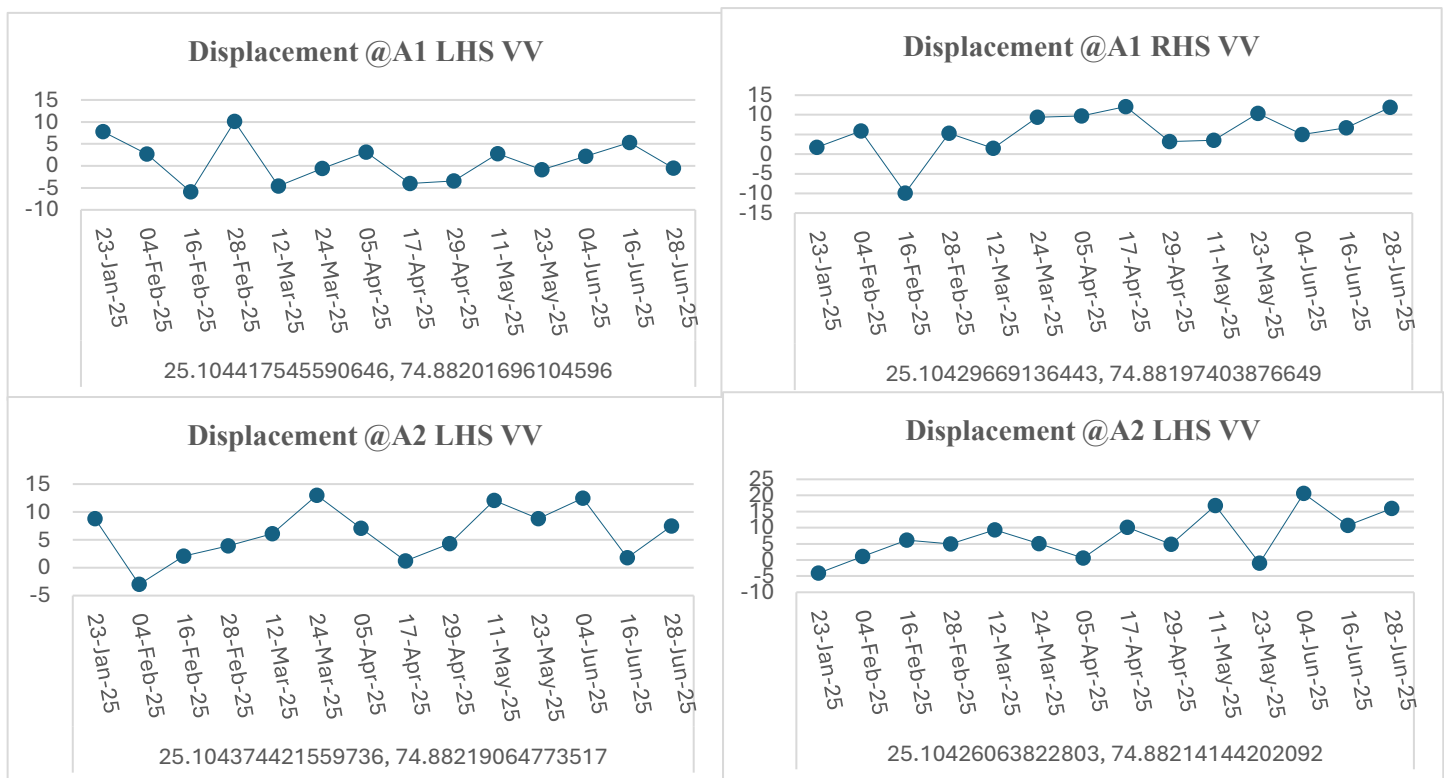


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

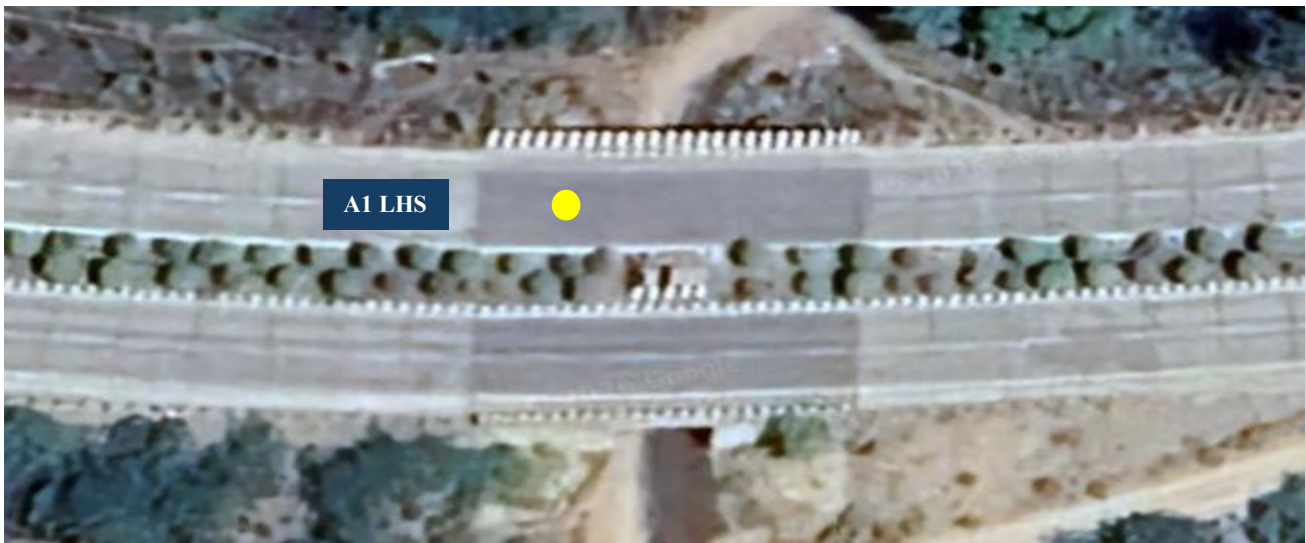
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



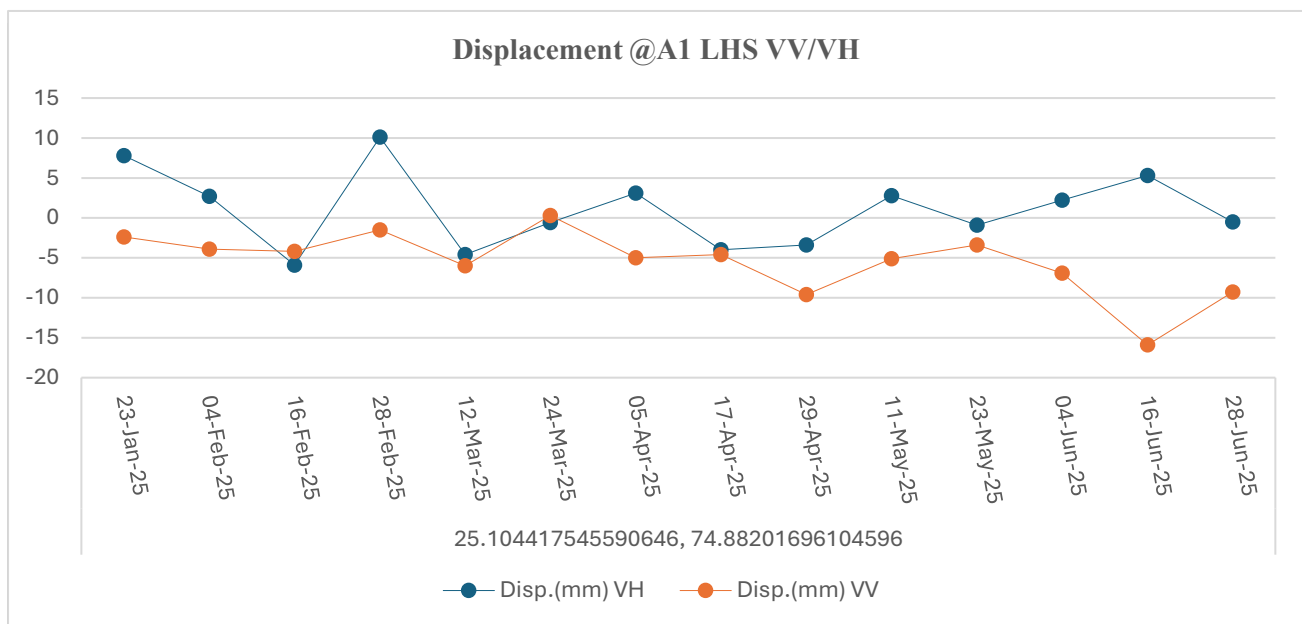
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.104417545590646, 74.88201696104596)



Key Observations:

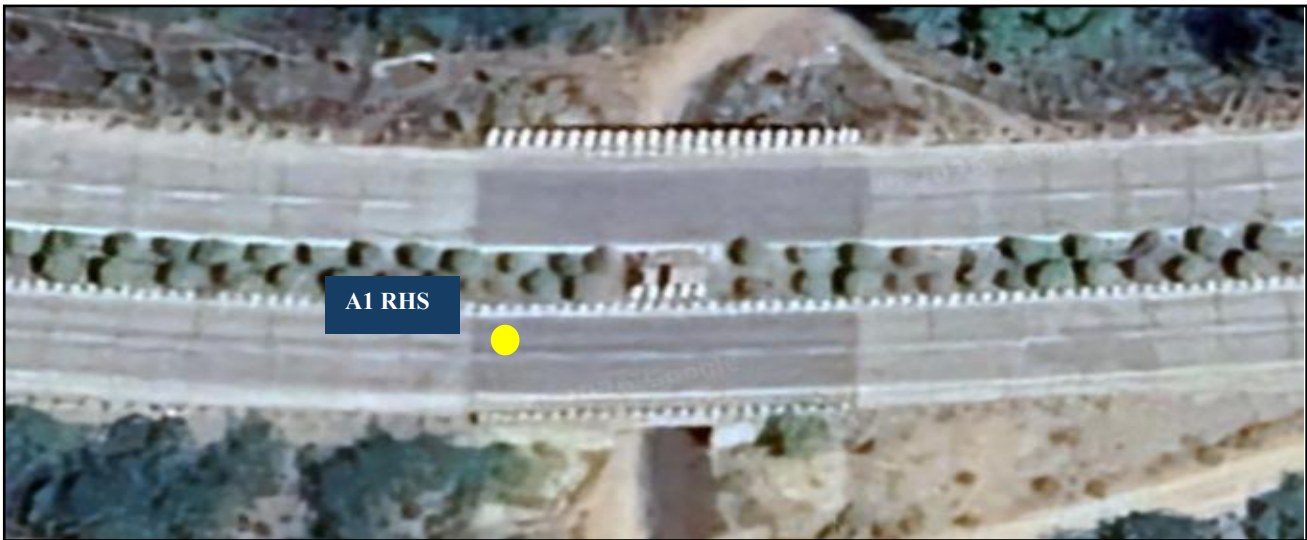
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** exhibits a predominantly negative trend, varying roughly from -2 mm to -15 mm.
- **Horizontal Displacement (VH)** shows moderate oscillatory behaviour, ranging approximately from -6 mm to +10 mm.

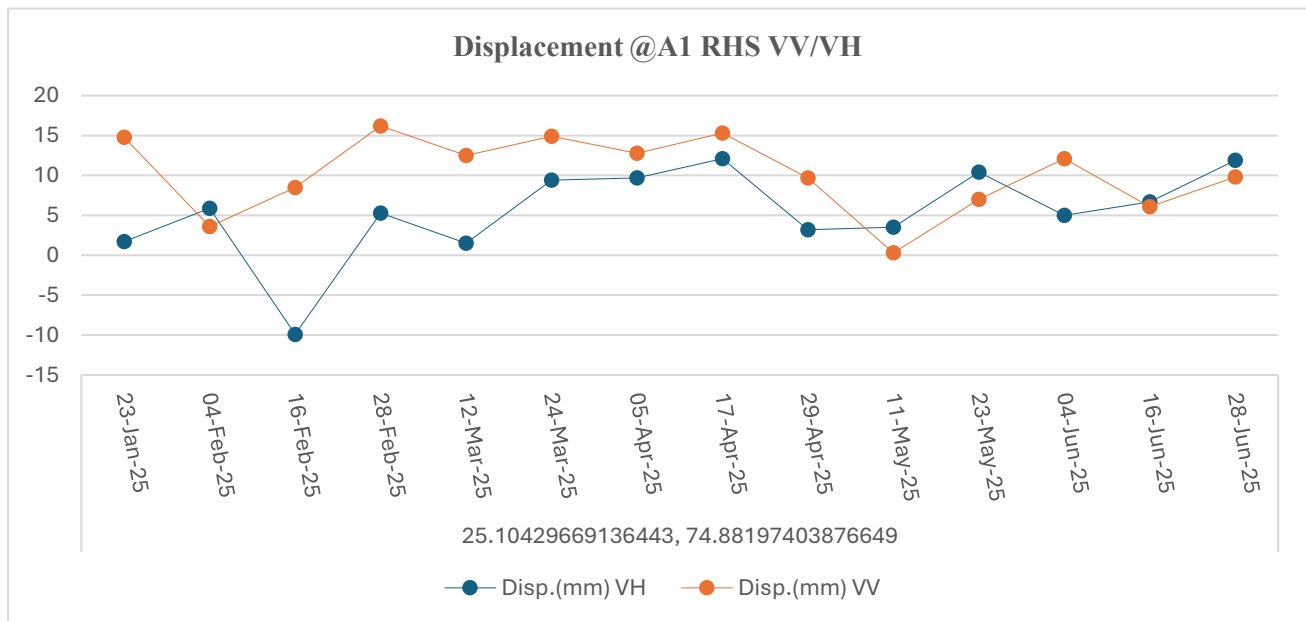
Critical Events & Anomalies:

- **January-February 2025:** VH remains positive initially (around +7 to +3 mm) while VV is mildly negative (-2 to -4 mm), suggesting normal operational response.
- **April 2025:** VV reaches a pronounced negative peak (~-10 mm) as VH remains near neutral, highlighting dominant vertical settlement.
- **June 2025:** VV records its maximum settlement (-15 mm) while VH stays moderately positive (around +2 to +5 mm), suggesting stabilized lateral behaviour with continued vertical settlement.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.10429669136443, 74.88197403876649)



Key Observations:

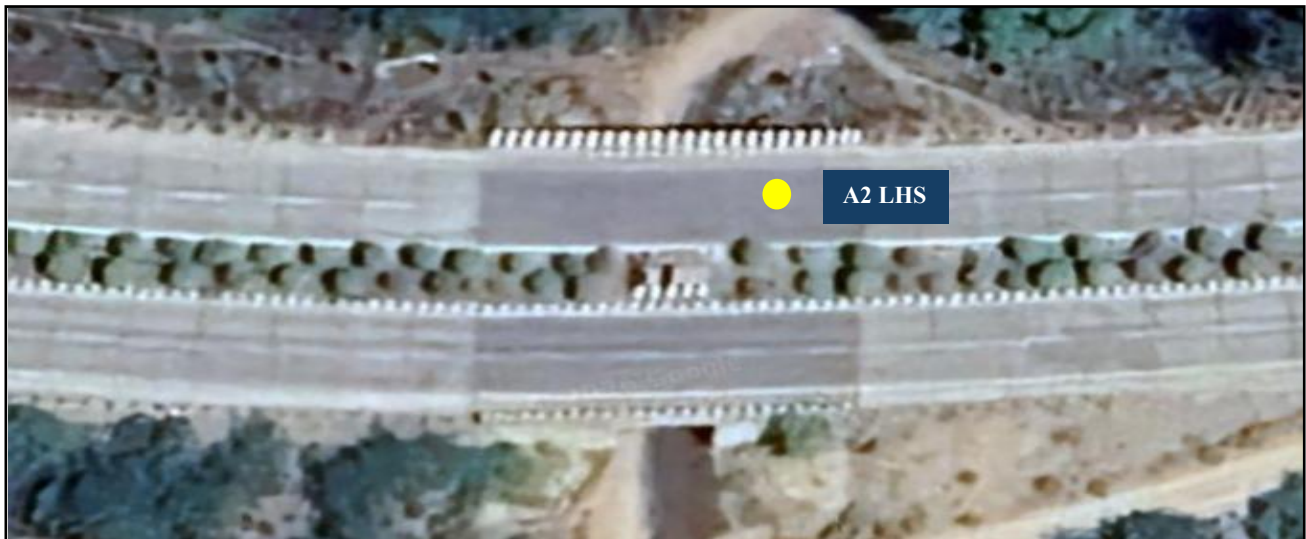
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH exhibits a moderate fluctuating trend, ranging approximately from -10 mm to +12 mm.
- **Vertical displacement (VV)** VV shows a predominantly positive trend, varying roughly from 0 mm to +16 mm.

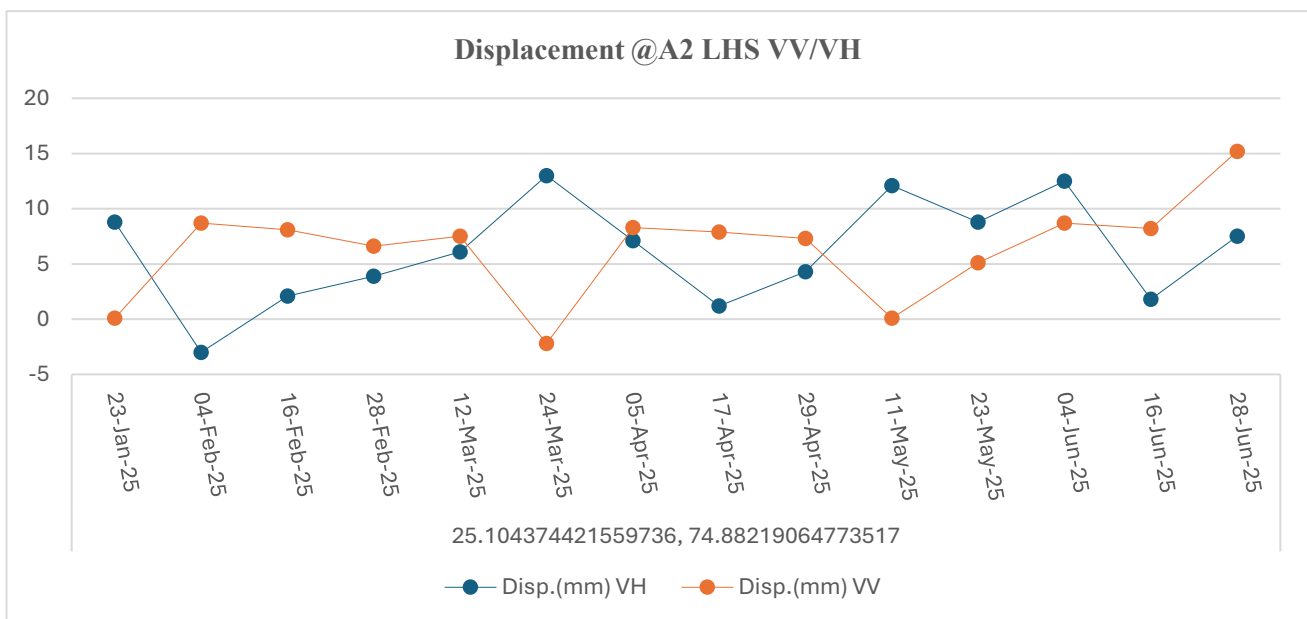
Critical Events & Anomalies:

- **January 2025:** VH remains slightly positive (+2 mm) while VV is relatively high (+14 mm), indicating initial vertical uplift with minimal lateral movement.
- **March 2025:** VH recovers to positive values while VV stays consistently high, indicating lateral realignment and stable uplift response.
- **May-June 2025:** Both VH and VV stabilize at moderate positive values (VH +5 to +12 mm, VV +6 to +12 mm), suggesting a balanced and stable displacement regime.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.104374421559736, 74.88219064773517)



Key Observations:

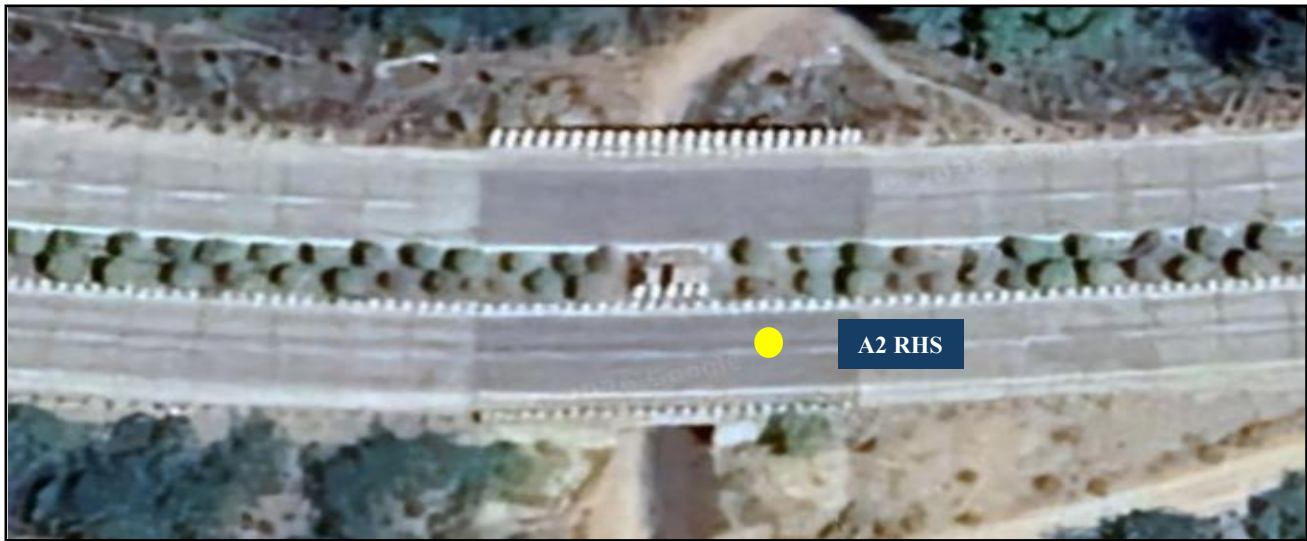
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate variability with repeated oscillations, ranging roughly between –3 mm and +13 mm.
- **Vertical Displacement (VV)** Vertical displacement at A2 LHS shows alternating uplift and settlement behaviour, with values ranging approximately from –2 mm to +15 mm uplift.

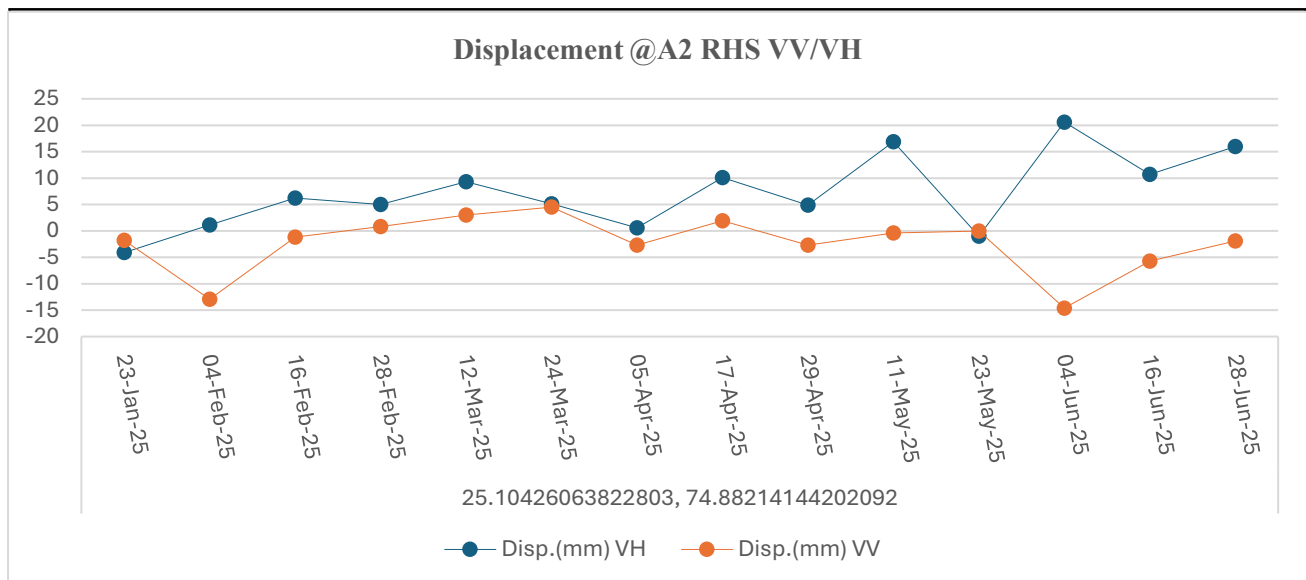
Critical Events & Anomalies:

- **January 2025:** VH begins at +8 mm, while VV is near 0 mm, indicating initial lateral displacement with negligible vertical response.
- **June 2025:** VV rises steadily to +15 mm, while VH fluctuates between +2 mm and +7 mm, indicating dominant vertical uplift toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.10426063822803, 74.88214144202092)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement at A2 RHS is predominantly negative, indicating settlement-dominated behaviour over most of the monitoring period.
- **Horizontal (VH) displacement** Horizontal displacement exhibits a strongly positive and progressively increasing trend, ranging roughly from 0 mm to +22 mm.

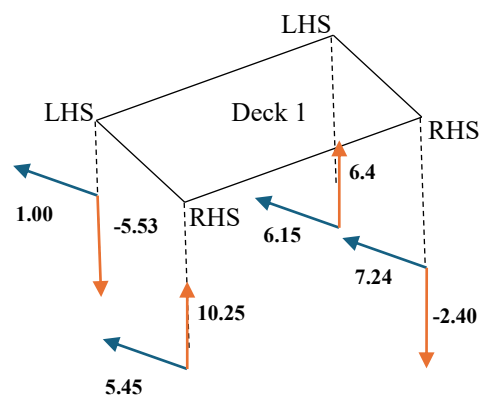
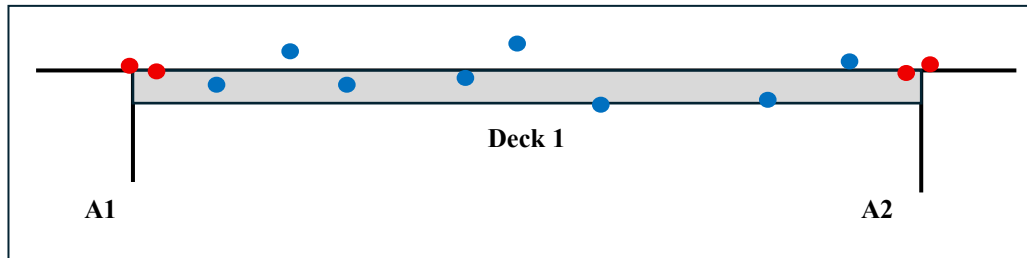
Critical Events & Anomalies:

- **January 2025:** VH begins near 0 mm, while VV is slightly negative (–1 mm), indicating initial settlement with minimal lateral displacement.
- **May 2025:** Both VH and VV indicate intensifying lateral movement accompanied by renewed settlement.
- **June 2025:** VH reduces slightly to +15 mm, while VV remains negative (–2 mm), confirming no full vertical recovery by the end of monitoring.

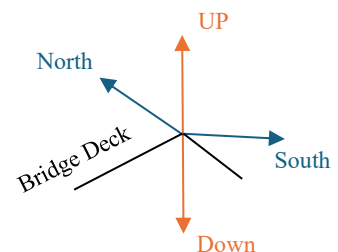
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab demonstrates a mixed vertical response, with both uplift and localized settlement observed across LHS and RHS supports. At the deck slab level near RHS, a moderate uplift of approximately +6.4 mm is observed, suggesting upward movement of the slab relative to the supports.

Horizontal Displacement Behaviour:

- Horizontal displacements display clear directional variability, with lateral movements occurring in opposing directions between LHS and RHS regions at the central RHS lower support, a horizontal displacement of approximately +6.15 mm is observed, indicating increased lateral movement relative to the LHS.

CONCLUSION:

- The bridge deck slab exhibits a non-symmetric deformation pattern, characterized by uplift-dominated behaviour near the RHS central support and localized settlement on the LHS. Overall displacement magnitudes remain within acceptable operational limits, with no immediate indication of structural distress or instability. Continued epoch-wise monitoring is recommended to confirm whether the observed uplift and lateral asymmetry are transient (thermal/load-related) or progressive in nature.